Dang Hyeona

Backend developer turning operational needs into product features

I build and operate Java/Spring Boot servers, turning complex requirements into API, data, and shared product capabilities.

Java · Spring Boot · Backend · Operation

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Development flow through work experience

A path from research and education to production development focused on turning operational needs into product capabilities

2019 — Summer intern · research trainee

ETRI Research Intern

Frame-editing UI and background-removal experiment flow for an AI-based video editing project

Tech	Python 3.6, Tkinter, OpenCV, GrabCut Pillow, NumPy, Keras, scikit-learn
Role	Video editing tool development 2-person project team

Work

- Built a Tkinter-based frame editing UI
- Implemented GrabCut-based manual background removal
- Experimented with CNN-based automatic background removal
- Adjusted learning layers and input sizes
- Configured a flow where users can select experiment conditions

Outcome

- Compared manual editing and automatic detection results using the F-measure
- Connected the experiment to a tool flow where users can select conditions and review results

2021-22 — Contract freelance

SSAFY Project Practice Coach

Student project coaching · coach operations standards

Tech	Project and presentation coaching, Git, Linux/CLI deployment Server operation support, operation documentation
Role	Project practice coach for SSAFY 6th-7th cohorts New-coach OJT and work guidance

Work

- Coached 66 teams from planning through presentation across six project cycles
- Supported coaching and operations for Samsung Electronics-linked projects
- Organized reporting templates, document workflows, and best-case samples for practice coaches
- Led about 15 live development sessions and formalized the workflow as an operation manual
- Operated open-source management and Twin Campus Seoul responsibilities

Outcome

- Defined project operation scope and new-coach work standards for SSAFY 6th-7th cohorts
- Organized external-linked projects, live sessions, and operation documentation into formal processes

2024-NOW — Full-time · senior

Bigxdata Data Solutions Team / Senior

Expanded an initial BI portal into a data portal product while building product capabilities and operational foundations.

Tech	Java 8-17, Spring Boot 2.x-3.4.5, JPA/MyBatis/QueryDSL PostgreSQL/Oracle/MariaDB/MySQL, Swagger
Role	Product features, shared management, and data catalog ownership Product structure transition and diverse operating environments

Work

- Built a DBMS management feature end to end, from planning and design through development and testing.
- Supported PostgreSQL and Oracle with Flyway-based database change management.
- Built shared email delivery and template features while reviewing impact across diverse operating environments.
- Covered patch-release design, task assignment, and implementation for about three to four months.
- Implemented visualization and DBMS features during the product structure transition, then expanded ownership into shared management, data catalog, and curation features.

Outcome

- Connected DBMS management capabilities to product features usable across diverse operating environments.
- Contributed to a product structure transition that improved feature extensibility.
- Built a foundation for data portal expansion through shared management and catalog capabilities.

Technical Skills

Technologies organized by usage level across work and project experience.

USAGE LEVEL

1 bar	2 bars	3 bars	4 bars	5 bars
Basic build	Feature build	Project	Work use	Improve/Ops

LANGUAGE
Java

Built Java 8-17 server logic and verified runtime errors and response model impact.

Java 8-17 · Server Logic · Debugging

FRAMEWORK
Spring Boot

Developed Spring Boot 2-3.4.5 product features with API, data access, and WAS structure in mind.

Boot 2-3.4 · JPA/MyBatis · Tomcat

API DESIGN
REST API

Handled API design through implementation, including auth and environment-based API flows.

Auth Branch · Condition API · Swagger

DATABASE
DBMS

Worked on service DB design, migration, and query improvement with PostgreSQL, MariaDB, and MySQL.

PostgreSQL · Migration · Index

WEB SERVER
Nginx

Reviewed server version upgrades and security header settings for service operations.

Version Response · Security Headers · Ops Check

CLOUD
AWS

Worked with load balancing, RDS, and monitoring setup for development and operations environments.

Load Balancing · RDS · Monitoring

VERSION CONTROL
Git

Created and applied team conventions while adapting to established production conventions.

Convention · Commit Scope · Push

COLLABORATION TOOL
Atlassian

Structured Jira and Confluence information and organized recurring team-level shared updates.

Jira · Confluence · Team Share

DEVELOPMENT SUPPORT
AI-assisted Development

Organized shared AI setup and skill structure that became an actual team convention.

Codex · Claude · Documentation

PRODUCT SCALING EXPERIENCE

PRODUCT
Productization Experience

Participated in expanding an existing project into a productized portal service by organizing feature flows.

Productized Service · Multiple Environments · Productization

ARCH
Scalable Structure Transition

Developed an extensible structure by considering feature boundaries and shared flows during a product structure transition.

Product Scaling · Structure Transition · Shared Area

COMMON
Shared & Hub Feature Contribution

Expanded my ownership from specific feature modules into shared areas and hub-like features.

Shared Feature · Hub-like Feature · Extension

SYSTEM
Team Development System

Organized AI usage flows as a team convention and continuously managed Confluence knowledge and onboarding materials.

AI Convention · Confluence · Onboarding Materials

Safers

A service that delivers environmental information through interactive experiences.

Period

Oct 12–Nov 19, 2021 (6 weeks)

Domain

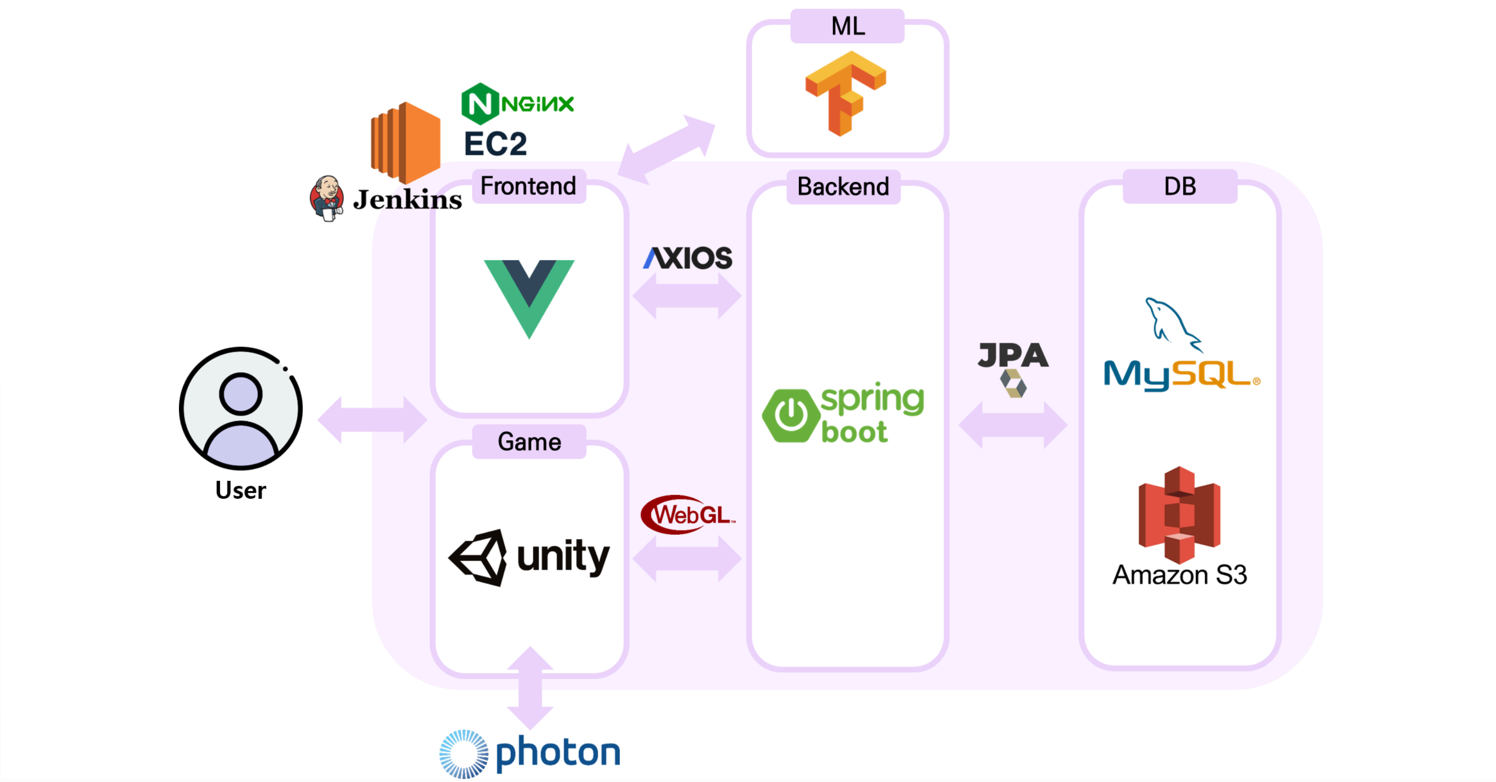
Web · Metaverse

Team

Five-person team · Team lead

Role

Backend · Infrastructure · ML · Presentation



Technologies and versions

The main technologies and runtime environment used for the project.

Language	Java 8
Backend	Spring Boot 2.5.6 · JPA
Web · platform	Vue 2.6.14 · Unity 2019.4.32f1 · Photon 2.38
Data · infra	MySQL 8.0.22 · AWS EC2/S3 · Jenkins 2.303.2 · nginx 1.18.0
ML	Teachable Machine Image 0.8.5 ·

Key responsibilities

Backend · API · Deployment · ML

- Designed data structures for users, animals, maps, missions, and logs
- Contributed to the design and implementation of core APIs for users, boards, and Unity integration
- Connected Unity WebGL with server data
- Configured the service environment on AWS EC2 and S3
- Built the Jenkins and nginx CI/CD flow
- Applied ML-based image mission validation

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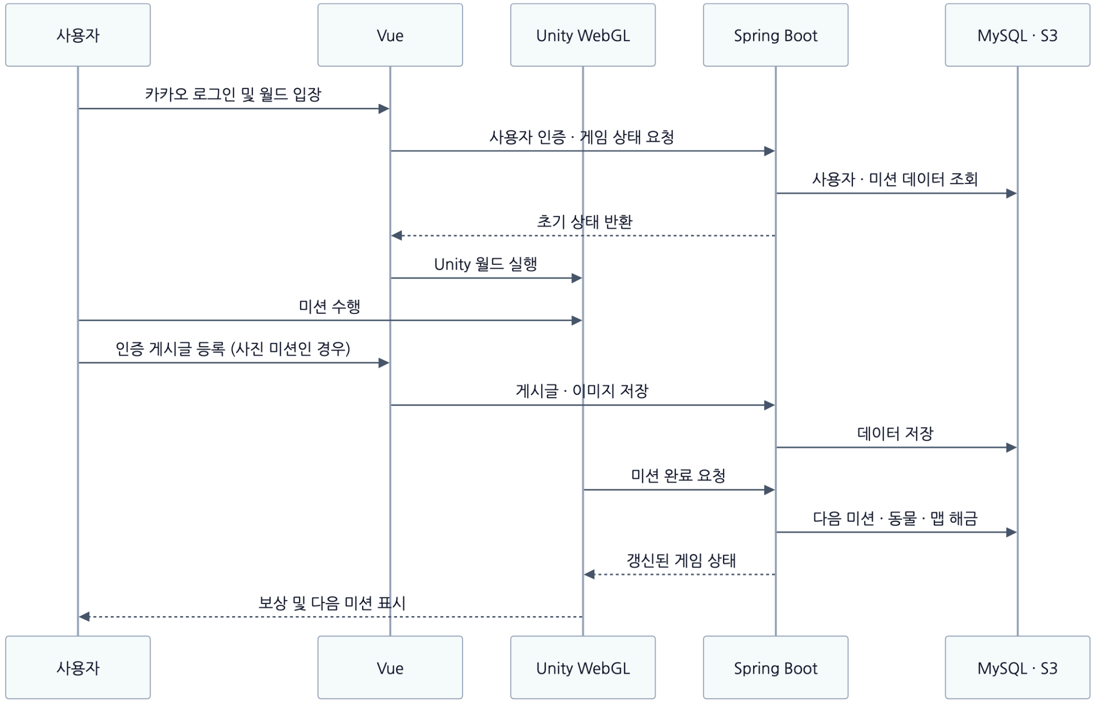
API design across the service flow



01 Shared code and content data design

Designed a structure that supports future service expansion.

- Designed four shared-code groups and fourteen detail codes
- Defined data and initial values for three maps, thirty-two animals, and thirty missions
- Connected state values and next-mission relations to the API flow

02 Unity WebGL and server integration design

- Connected the user-authentication and initial-state flow after login
- Integrated user, mission, and animal data between Vue, Unity WebGL, and Spring Boot APIs
- Structured the API flow from mission-result storage to next-content state updates

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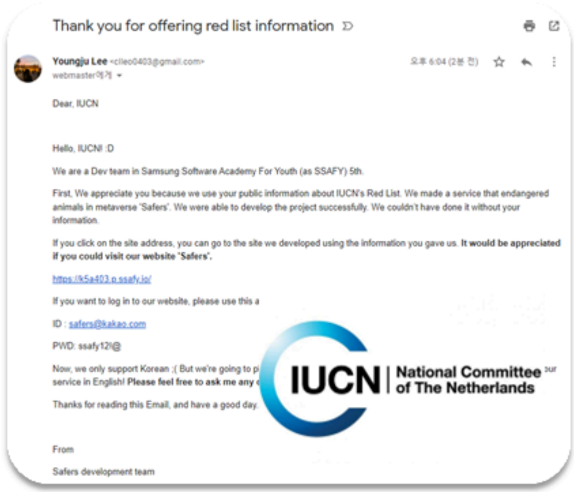
Outcome and external validation



01 Operational foundation

- Resolved Jenkins·SSH issues and established an automated deployment flow
- Documented the reproduction, fix, and redeployment flow for deployment-version errors
- Documented deployment response procedures for repeated team use

02 External validation and project result

- Confirmed a positive response from the National Institute of Ecology
- Communicated externally about using IUCN Red List information
- Placed first among eight teams in the SSAFY 5th autonomous project
- Won nationally as one of the top eight teams across four campuses and 750 participants

Node-RED Motion Pose

A public project that made camera-based motion recognition usable in Node-RED and SmartThings environments.

Period

Aug 23–Oct 8, 2021 (7 weeks)

Domain

AI · IoT · Node-RED

Team

5-person team · Team lead

Role

Node-RED development · Presentation · Distribution

Technologies and versions

The technologies used for motion recognition and Node-RED distribution.

Runtime	Node.js 14.17.3 · Node-RED 2.0.6
Recognition	MediaPipe BlazePose · Hands
Communication · server	Socket.IO 4.2.0 · Express 4.17.1
Distribution	npm package 1.1.3 · Node-RED Library

Key responsibilities

Hand recognition · Node-RED · Distribution · Contribution

- Built the hand-recognition flow for the Hands Pose Node
- Implemented hand registration and result discovery
- Packaged the features as reusable Node-RED nodes
- Published nine nodes to npm and the Node-RED Library
- Contributed PRs #46 and #47 to Samsung Automation Studio

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A public project that made camera-based motion recognition usable in Node-RED and SmartThings environments.

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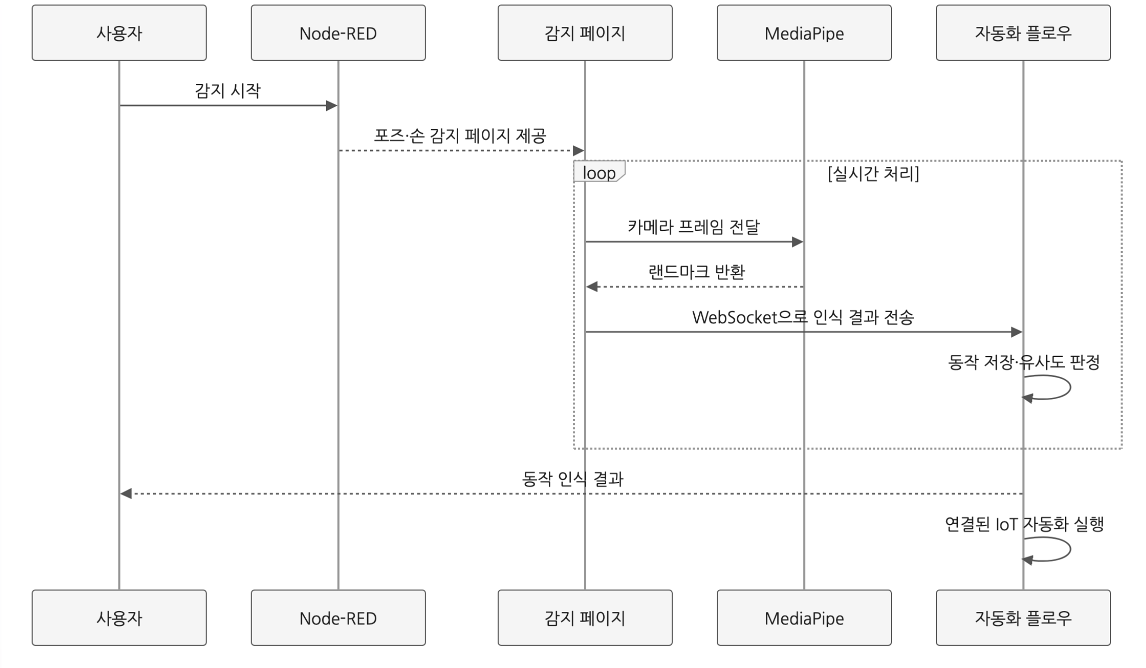
From input to result: motion-recognition flow



01 Camera input · landmark extraction

- Separated webcam and SmartThings camera inputs with pose-detect-webcam and pose-detect-iotcam
- Extracted body and hand landmarks with MediaPipe BlazePose and Hands
- Passed real-time frame results to the next recognition step

02 Registration · discovery · Node-RED integration

- Saved motion data and searched for similarity with pose-register and pose-find
- Configured hand registration and discovery with hand-register and hand-find
- Delivered recognition results to the automation flow through monitor and WebSocket
- Modularized detection, registration, and discovery for composition in Node-RED

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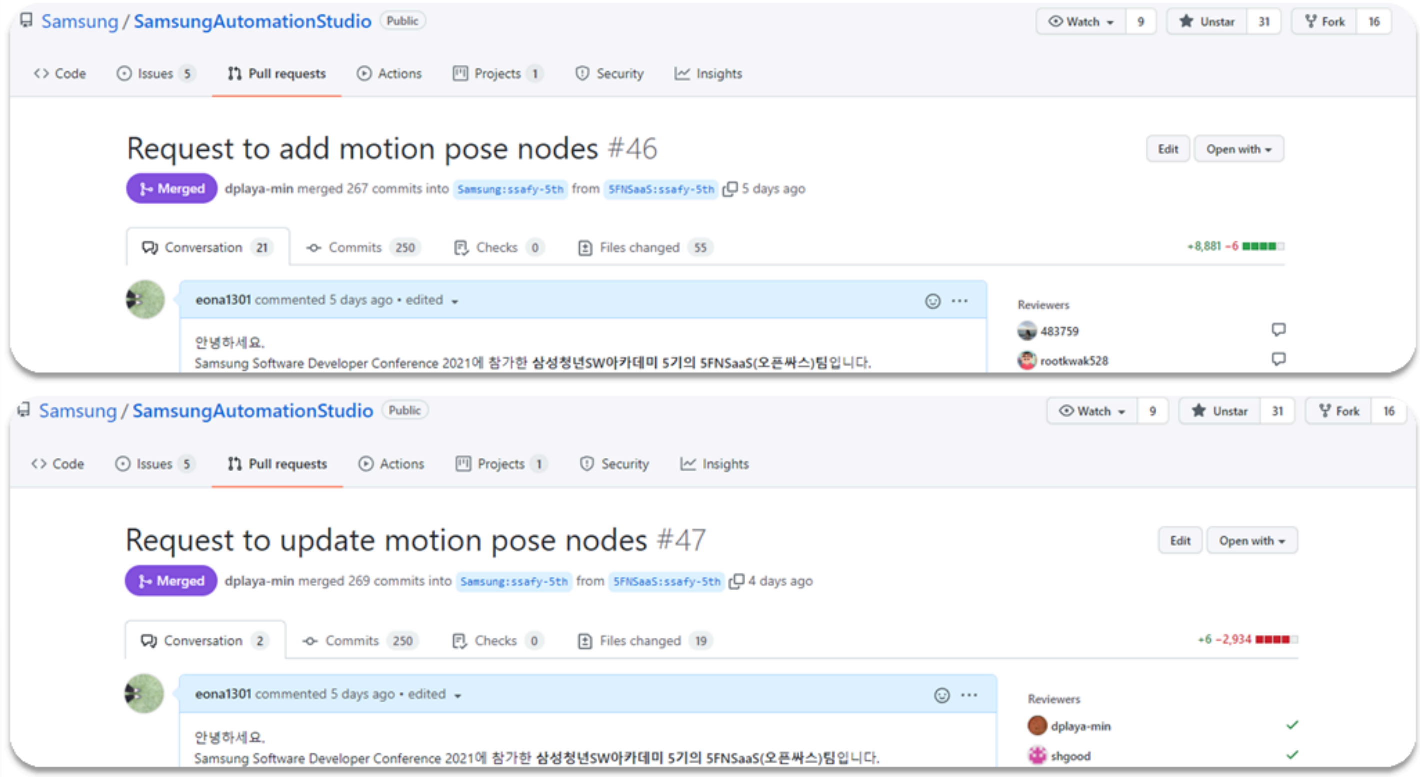
Open-source release · contribution · presentation



01 npm · Node-RED Library release

- Published nine motion-recognition nodes as npm package 1.1.3
- Registered the package in the Node-RED Library and supported webcam and SmartThings cameras
- Recorded more than 600 downloads in one week

02 Open-source contribution and presentation

- Merged PRs #46 and #47 into Samsung Automation Studio
- Received 10+ code review comments on public PRs and refined the implementation
- Presented the release, open-source contribution, and IoT use case flow at SSDC 2021 Share

Development Practice and AI

I use Codex and Claude in the development flow, applying changes only after team standards and verification.

AI-assisted decision flow

01 → **Spot recurring work**
Identify a reusable work pattern

02 → **Review team usage**
Understand current use and needs

03 → **Agree on scope**
Align criteria through review and discussion

04 → **Build shared setup**
Configure setup, skills, and guidance

05 → **Verify in practice**
Apply to recurring work and check results

06 **Document and share**
Turn it into reusable team guidance

01 Team Standards

Shared AI Environment

01 **Structuring shared setup and skills**
Building the base environment and reusable units

02 **Documenting team usage criteria**
Defining usage principles around context and security

02 Verification and Ownership

Verify Before Applying

01 **Checking tests, builds, and diffs**
Verifying behavior and impact through execution

02 **Comparing docs with project code**
Checking fit with the actual version and environment

03 Team Adoption and Scale

Making Recurring Work Shared

01 **Sharing scope in team discussions**
Defining the shared scope from use cases and feedback

02 **Preparing Confluence and onboarding materials**
Writing criteria and examples for continued reuse

Key outcome **Individual use → Shared team practice** Turned shared setup, skills, and usage criteria into a reusable development environment for recurring work.